A square product for six values

Problèmes ouverts — A. Dujella · 2026

Problem 1.5 (Dujella, 2026). Is there any triple a_1, a_2, a_3 of distinct positive integers such that $(a_1x+1)(a_2x+1)(a_3x+1)$ is a perfect square for at least six positive integers x ?

Remark. An example by Petričević (2023) shows that $(x+1)(55x+1)(276x+1)$ is a perfect square for at least the following five positive integers : $x = 4, 8, 17, 89, 4870844$. Other known examples with at least five solutions are $(a_1, a_2, a_3) = (1, 2, 53)$, $(1, 5, 26)$, $(1, 10, 1407)$, $(1, 11, 316075)$, $(1, 24, 97)$, $(1, 84, 10333)$, $(5, 24, 3749)$, $(1, 571, 27409)$ (Petričević (2023, 2026)), $(1, 8, 28440881)$ (Dujella (2026)), $(3, 19, 7858099)$ (Sadowski (2026)). It is not known whether there exist infinitely many such examples.

There are certain infinite families with at least four positive integer solutions, including those with $(a_1, a_2) = (1, 2)$ (Sadowski (2026)), $(a_1, a_2) = (1, 8)$, $(a_1, a_2) = (k^2 - 2, k^2 - 1)$ for $k \geqslant 2$, $(a_1, a_2, a_3) = (a, 3a + 1, 24a^2 + 12a + 1)$, where $6a + 1 = p^2$ and $3p^2 + 5 = 2q^2$ (Dujella (2026)), $(a_1, a_2, a_3) = (1, k^2 + 2, k^8 + 4k^6 + 5k^4 + k^2 - 1)$ (Sadowski (2026)), $(a_1, a_2, a_3) = (1, k^4 - 3k^2 + 1, k^6 - 7k^4 + 13k^2 - 3)$ (Andrašek (2026)).

The first infinite family with five solutions was found by Christopher Sadowski (2026) : $a_1 = 1$, $a_2 = 12t+7$, $a_3 = 12t^2+19t+8$ with $x = 1$, $x = (21t+13-3s)/2$, $x = 4t+1$, $x = (21t+13+3s)/2$, $x = (16t^2 + 20t + 5)(144t^3 + 276t^2 + 157t + 24)$, where $s^2 = 33t^2 + 46t + 17$. From $(t, s) = (4, 27)$ and $(t, s) = (47, 274)$, we recover the previously known examples with five solutions $(1, 55, 276)$ and $(1, 571, 27409)$, while the next pair $(t, s) = (9932, 57059)$ gives $(a_1, a_2, a_3) = (1, 119191, 1183924204)$ with five solutions $x = 9932, 18704, 39729, 189881, 22274321918202550722995 6$.